# Sahaj Bhadja

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## Summary

Spring 2026 M.S. CS graduate with 2+ years building **Python backends**, **React/TypeScript** interfaces, and **AWS** services for scalable, data-driven products. Passionate about software craftsmanship, marketplace problems, and performant user experiences.

## Education

**University of Southern California, Los Angeles, CA** **August 2024 – May 2026**
*Master of Science in Computer Science* *GPA: 3.75/4.0*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Software Engineering

**Pandit Deendayal Energy University, Gandhinagar, India** **November 2020 – May 2024**
*Bachelor of Technology in Computer Science and Engineering* *GPA: 3.88/4.0*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Structures

## Experience

**USC Dornsife** **Los Angeles, CA**
*Software Engineer* *September 2025 – Present*

- Built **Python/FastAPI** backend services and **PostgreSQL** data layers for 500+ users on cloud infrastructure, reducing dashboard latency from **1.2s to 350ms** while supporting data-driven product analytics across multi-tenant platform workflows.
- Designed ETL-style pipelines and automated validation for **5K+ monthly jobs** with **99%+ reliability**, enabling scalable releases, faster incident triage, and continuous experimentation on production integration workflows in fast-paced delivery cycles.

**Grad8** **Los Angeles, CA**
*Software Engineer Intern* *June 2025 – August 2025*

- Developed full-stack features with **Python/Flask**, **React**, and **TypeScript** handling **10K+ daily requests**, translating stakeholder needs into performant APIs and inclusive user interfaces while maintaining high standards for software craftsmanship and product quality.
- Shipped AI-assisted product workflows for **150+ instructors** reducing feedback turnaround by **60%**, collaborating in Agile sprints, running iterative experiments, and learning from senior engineers through code reviews across cross-functional teams.

**USC AD Labs** **Los Angeles, CA**
*Software Engineering Assistant* *January 2025 – May 2025*

- Built experimentation pipelines across **100K+ records**, enabling reproducible evaluation and data-driven decisions that improved closed-loop stability for complex systems with inventory and matching-style technical challenges.
- Partnered with researchers on solution design and algorithm experiments, communicating findings across teams and strengthening engineering quality as prototypes matured toward production-ready performance improvements.

**Aasma Technology Solutions** **India**
*Software Developer Engineer - (SecureSync)* *August 2023 – July 2024*

- Engineered **AWS**-backed API and ETL microservices with **Docker** deployments improving response time by **30%**, supporting distributed integrations, inventory-style data sync, and stable delivery across high-volume production marketplace workloads at scale.
- Maintained **20+ automated test suites** in CI pipelines using PyTest and Git workflows, increasing release confidence by **68%** through rigorous code reviews, debugging practices, and continuous deployment across Windows and Linux production environments.

**Aasma Technology Solutions** **India**
*Full Stack Developer Intern - (VibrantChem)* *May 2023 – August 2023*

- Developed **MERN** platform with **WebSockets** and REST APIs syncing **2M+ records** across locations with **33% fewer conflicts**, delivering real-time performant dashboards for multi-sided operational workflows in a dynamic startup environment.
- Automated **Docker**-based build and deployment flows with Jenkins, cutting release time by **50%** while scaling software delivery practices and holding frontend and backend code to meticulous quality standards under deadline pressure.

## Skills

**Languages**: **Python**, **TypeScript**, JavaScript, Go (familiar), C# (familiar), Java, SQL, Bash
**Frontend**: **React**, Next.js, HTML5, CSS3, responsive design, inclusive UI, component-driven development
**Backend**: FastAPI, Flask, Node.js, REST APIs, microservices, WebSockets, software craftsmanship, product mindset
**Data & Cloud**: **PostgreSQL**, MySQL, MongoDB, **Redis**, Elasticsearch (familiar), **AWS**, Docker, Kubernetes, Git, GitLab (familiar), CI/CD, Agile, PyTest, distributed systems, marketplace systems, real-time updates

## Projects

**FinNexus** Live | *TypeScript, PostgreSQL, WebSockets, Kubernetes*

- Built fintech marketplace supporting **50K+ users** and **1M+ transactions** with Plaid, PostgreSQL, real-time WebSocket inventory updates, and Kubernetes deployment for performant many-sided buyer-seller experiences at scale.

**PiXMod** Live | *Next.js, TypeScript, MongoDB, Stripe*

- Shipped AI-powered Next.js SaaS with Stripe billing, MongoDB workflows, and polished responsive interfaces, improving premium conversion by **18%** through performant inclusive UI patterns that delight users and enhance product brand perception.

**Big-Brain** Live | *Python, PostgreSQL, LLM, Semantic Search*

- Developed data-driven document platform with PostgreSQL, semantic search, and real-time collaboration for **200+ users** achieving **95% query accuracy**, empowering decision-making through intuitive interfaces and scalable cloud-backed analytics workflows.